

Classical CV Pipeline — Stage-by-Stage Output

1. Input frame



RGB camera image (1280 × 720)

2. BEV warp



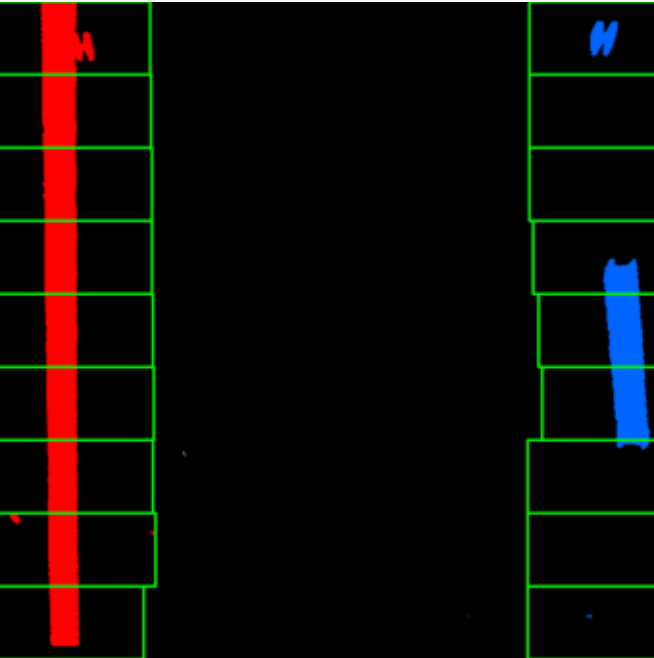
Perspective transform to bird's-eye view (720 × 720)

3. Binary threshold (in BEV)



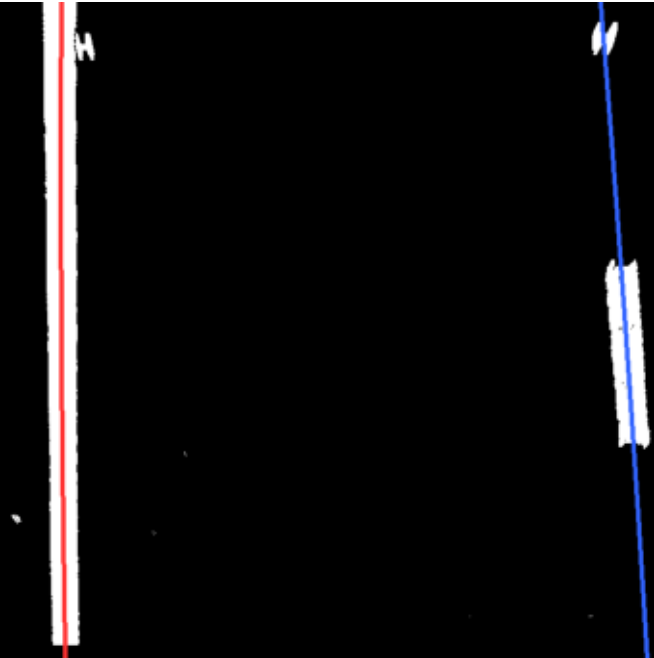
Sobel x-gradient ∪ HLS color → binary lane mask

4. Sliding window



Histogram peak → 9 windows track lane pixels (red = left, blue = right)

5. Polynomial fit



Least-squares 2nd-degree polynomial $x = ay^2 + by + c$

6. Final overlay



Curves unwrapped back to camera view + drivable polygon